

Mohit Singh Chouhan

AI/ML Engineer • Generative AI & RAG Systems • Data Science

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Professional Summary

AI/ML Engineer with hands-on experience building Machine Learning and Generative AI pipelines, now applying that foundation to RAG systems, autonomous AI agents, and NLP workflows. Skilled in Python, REST API development, and end-to-end model deployment – turning data preprocessing, training, and evaluation work into production-ready, business-facing applications.

Experience

Samyak Infotech **Apr 2026 – Jun 2026** *Data Science Trainer* *Bhilwara, India*

- Trained learners in Python, Machine Learning, and Deep Learning fundamentals through structured, hands-on sessions.
- Designed practical curriculum and coding exercises enabling students to build and evaluate real-world ML/DL models independently.

Airocode Technologies **Jan 2026 – Apr 2026** *Full Stack AI/ML Engineer – Trainee* *Pune, India*

- Built and deployed production-grade ML pipelines in Python – preprocessing, feature engineering, cross-validation, and Flask-based deployment – for scalable data-driven applications.
- Integrated model outputs with REST APIs and relational databases, enabling real-time predictions and structured reporting for business use cases.

iamneo (An NIIT Venture) **May 2025 – Jun 2025** *Industry Training Trainee – DSA & DBMS* *Parul University*

- Completed intensive training in Data Structures, Algorithms, and DBMS, reinforcing query optimization for large-scale data workflows.

Skills

Generative AI & LLMs	Prompt Engineering, RAG, LLM APIs (Claude, OpenAI, Groq), Tool/Function Calling, AI Agents (ReAct), Agent Memory, LangChain
Machine Learning	Regression, Classification, Clustering, Ensemble Methods (Random Forest, XGBoost), SMOTE, Recommendation Systems
NLP & Vector Search	TF-IDF, Cosine Similarity, Sentence-Transformers, ChromaDB, FAISS, Semantic Chunking, Content/Collaborative Filtering
Programming	Python, SQL, Java, REST API Development (Flask)
Frameworks & Libraries	Scikit-learn, Pandas, NumPy, PyTorch, TensorFlow, Flask, FastAPI, Streamlit, SQLAlchemy, Matplotlib, Seaborn
Databases & Tools	MySQL, MongoDB, ChromaDB, Git, Power BI, Jupyter, Vercel, Render
Core Competencies	Data Cleaning & Preprocessing, Feature Engineering, EDA, Model Evaluation & Deployment, DSA (500+ LeetCode)

Projects

Chat With Any PDF – RAG-Powered Document Q&A | *Python, ChromaDB, Sentence-Transformers, Groq API* [GitHub](#)

Built a RAG pipeline with PDF extraction, token-based chunking, local embeddings, and ChromaDB cosine-similarity retrieval, eliminating hallucination via page-level source citations.

Personal Research Agent – Autonomous ReAct AI Agent | *Python, Groq API, ChromaDB* [GitHub](#)

Designed an autonomous ReAct agent that plans web searches, gathers evidence via tool calling, and writes cited research reports with persistent memory and step-count guardrails.

Hybrid Financial News Recommendation System | *Python, Flask, NLP, TF-IDF* [GitHub](#)

Built a hybrid content-based and collaborative filtering recommendation engine achieving **25% accuracy improvement** over baseline, deployed as a Flask API.

Telco Customer Churn Predictor | *Python, XGBoost, SMOTE, Flask* [GitHub](#)

Built an end-to-end churn prediction system with SMOTE for class imbalance, deployed as a Flask REST API with a real-time Chart.js risk dashboard.

Education

Parul University **Vadodara, Gujarat**

B.Tech – Computer Science and Engineering

CGPA: 7.57/10

Jun 2022 – Jun 2026

Achievements & Certifications

- TCS iON NQT 2026** – Scored **75.88%** in national-level aptitude and technical assessment.
- Tata:** GenAI Data Analytics (Forage) | **IBM:** SQL & Relational Databases | **Microsoft:** Python Programming | **Oracle:** JDBC
- Vadodara Hackathon 5.0** – Top 45 Teams (Team InnoVentures) | **Parul Hack Verse Hackathon 2025**
- NPTEL:** IoT & Computer Networking | Solved **500+ LeetCode problems** (Easy, Medium, Hard)